

Table 3. Baseline results for the causal variable localization track. In each table, the first row is the task, the second row is the model, and the third row is the causal variable. For Arithmetic, MCQA, and ARC (Easy), we report interchange intervention accuracy, i.e., the proportion of aligned interventions on the causal model and deep learning model that result in the same output token(s); higher is better. For each method of aligning a causal variable to LM features, we report the mean across counterfactual datasets and layers in the low-level model. In parenthesis and bold, we report the best alignment across all layers. For IOI, we report the mean-squared error between the causal model logit and the deep learning model logit; lower is better. See App. F.5 for more detailed results by task.

ARC (Easy)					Arithmetic (+)		
Method	Gemma-2		Llama-3.1		Method	Gemma-2	Llama-3.1
	O_{Answer}	X_{Order}	O_{Answer}	X_{Order}		X_{Carry}	X_{Carry}
DAS	88 (94)	76 (88)	88 (99)	74 (84)	DAS	31 (35)	54 (65)
DBM	82 (99)	63 (80)	85 (100)	69 (82)	DBM	33 (43)	47 (58)
+PCA	78 (98)	64 (81)	84 (100)	72 (83)	+PCA	32 (44)	37 (56)
+SAE	70 (89)	54 (70)	74 (94)	55 (67)	+SAE	32 (44)	38 (55)
Full Vector	63 (100)	43 (74)	68 (100)	47 (72)	Full Vector	29 (35)	35 (45)

(a) The ARC (Easy) task with a high-level model that computes the ordering of the answer X_{Order} and then the answer token O_{Answer} .

(b) The two-digit arithmetic task with a variable that computes the carry-the-one variable X_{Carry} .

MCQA							IOI		
Method	Gemma-2		Llama-3.1		Qwen-2.5		Method	GPT-2	
	O_{Answer}	X_{Order}	O_{Answer}	X_{Order}	O_{Answer}	X_{Order}		S_{Pos}	S_{Tok}
DAS	95 (97)	77 (93)	94 (100)	77 (91)	86 (95)	78 (100)	DAS	2.20	2.08
DBM	84 (99)	63 (84)	86 (100)	66 (73)	46 (94)	60 (99)	DBM	2.22	2.35
+PCA	57 (96)	52 (81)	65 (99)	53 (74)	22 (76)	54 (100)	+PCA	2.24	2.33
+SAE	73 (90)	51 (65)	80 (99)	58 (65)	—	—	Full Vector	2.45	2.82
Full Vector	61 (100)	44 (77)	77 (100)	46 (68)	35 (99)	49 (99)			

(c) The MCQA task with variables for the ordering of the answer X_{Order} and then the answer token O_{Answer} . This is a low-data regime (≈ 100 examples).

(d) The IOI task with variables for the position S_{Pos} and token S_{Tok} of the subject. The metric is mean-squared error; lower is better.

RAVEL						
Method	Gemma-2			Llama-3.1		
	A_{Cont}	A_{Country}	A_{Lang}	A_{Cont}	A_{Country}	A_{Lang}
DAS	75 (85)	57 (67)	62 (70)	75 (83)	58 (64)	63 (70)
DBM	66 (71)	53 (65)	54 (58)	68 (80)	53 (59)	58 (64)
+PCA	63 (70)	47 (53)	50 (56)	62 (74)	48 (54)	53 (57)
+SAE	64 (72)	49 (56)	53 (59)	64 (72)	50 (57)	55 (57)
Full Vector	48 (62)	49 (57)	45 (56)	53 (62)	47 (53)	47 (57)

(e) The RAVEL task with variables for the country A_{Country} , continent A_{Cont} , and language A_{Lang} of a city.

Y_1 , and Y_{10} that represent the ones and tens digits of each two-digit input. The variable X_{Carry} is a child of X_1 and Y_1 and takes on the value 1 if $X_1 + Y_1$ is greater than 10. The output variable O_{110} has all inputs and X_{Carry} as parents and takes on the value $X_{\text{Carry}} + X_{10} + Y_{10}$. The output variable O_1 has X_1 and Y_1 as parents and takes on the value $(X_1 + Y_1) \% 10$. For the benchmark, we report results for localizing the variable X_{Carry} .

Counterfactual dataset. We use equal parts random counterfactuals and counterfactuals that do not change the carry-the-one variable (e.g., base $17+75$ and counterfactual $11+71$). Interchange interventions on X_{Carry} in $\mathcal{H}_+$ with random counterfactuals will cause the output variable O_{110} to

increase or decrease by 1 half the time and have no effect the other half. The carry-the-one counterfactual inputs always require a change in the output, but hold the input and parts of the output fixed so that low-level interchange interventions are less likely to have unintended consequences.

Results. For each baseline, we target the last token and the last token of the second operand. We see poor performance across the board (Table 3b; App. F.5.2). The Gemma-2 results are at chance and the Llama-3.1 results have only faint signs of success. The difference in results is likely due to only Llama-3.1 tokenizing multi-digit numbers as single tokens. There may be no “carry-the-one” variable present in these models. Alternatively, the variable might

be represented in a *non-linear features space*, e.g., an onion representation (Csordás et al., 2024). Future submissions that beat baselines will require genuine progress.

4.5. RAVEL

The Resolving Attribute–Value Entanglements in Language Models (RAVEL) benchmark (Huang et al., 2024a) evaluates methods for featuring hidden vectors and selecting features that isolate certain *attributes* about an *entity*. For causal variable localization track, we include the split of RAVEL for disentangling the country, continent, and language attributes of city entities. The prompts are queries about a certain attribute, e.g., *Paris is on the continent of*.

Causal model. The causal model for RAVEL has a text input variable T , three attribute variables $A_{Country}$, A_{Cont} , and A_{Lang} for the attributes of the city in the prompt, a queried attribute variable A_{Query} , and an output variable O that retrieves the value of the attribute variable corresponding to the queried attribute.

Counterfactuals. Half of the counterfactual prompts are prompts that query a different attribute from the base prompt. The other half are random sentences from Wikipedia containing the city. Interventions on $A_{Country}$, A_{Cont} , and A_{Lang} will only change the output if the queried attribute matches the intervened variable. When evaluating each variable, we balance the base prompts such that half of the prompts query the intervened attribute, which enforces the balance between interventions that should change the output and interventions that shouldn’t change the output.

Results. For each baseline, we target the last token and the last token of the city entity in the prompt. We generally see evidence that the attributes can be localized and disentangled (Table 3e; App. F.5.3), though the Llama-3 results are weaker. Because the dataset requires balancing causing an attribute to change with not changing the other attributes, the full vector baseline entirely fails completely.

4.6. Indirect Object Identification.

In App. A of Wang et al. (2023), there is an experiment where the identified four “S-Inhibition” heads that decrease the likelihood of the subject token are intervened upon with counterfactual datasets that invert the position of the subject or invert the token identity of the subject. They found the heads contain token and position signals that make roughly linear contribution to the difference between the indirect object logit and the subject logit. However, they did not attempt to disentangle the hidden vector outputs of those attention heads to align each signal with LM features.

Causal model. First, we replicate these experiments on our curated datasets and fit our own linear model that we use to define a causal model $\mathcal{H}_{IOI}$ that predicts the logit difference between the indirect object and the subject. The high-level causal model takes in a text input T and computes the token and positional information S_{Tok} and S_{Pos} encoding the subject token identity and position, respectively. Then, the output variable O : (i) checks if the token and position variables S_{Tok} and S_{Pos} match the input T and inverts the signal if a mismatch is detected, and (ii) computes the logit difference between the indirect object and subject as a linear function of these binary variables. See Appendix F.5.4 for details of the causal model.

Counterfactuals. We use counterfactuals where the subject and indirect object tokens are inverted, their position is inverted, or both. We align each variable with features of the four heads. These counterfactuals test the ability of methods to disentangle the token and position variables.

Results. Broadly, we find evidence that the position and token variable can be disentangled (Table 3d; App. F.5.4). For the full vector baseline, we conduct a brute force search and find an alignment of position to heads 7.3, 7.9 ($\Pi_{S_{Pos}}$), and 8.6 and token to head 8.10 ($\Pi_{S_{Tok}}$), to be better than other alignments to entire heads. While the variables can be disentangled at the level of heads, even better results are achieved when each variable is aligned to features of heads.

4.7. General Discussion

Table 3 shows the full results for the causal variable localization track. We now describe general trends across tasks.

Distributed alignment search (DAS) consistently achieves the best results. DAS is the only method that learns features with supervision from the high-level causal model, so it is not surprising this method performs best. This establishes a skyline to compare unsupervised featurization methods against.

DBM on standard hidden dimensions is a strong baseline. The success of DBM shows that the dimensions of untransformed hidden vectors can be useful units of analysis; however, DAS outperforming DBM shows that non-basis-aligned directions in activation space are generally better units of analysis.

DBM on PCA or SAE features does not outperform DBM on standard dimensions. SAE and PCA features generally fail to improve upon neurons, i.e., standard dimensions of hidden vectors, as a unit of analysis. Across the board, DBM on SAE features is the worst performing method other than the full vector baselines. A notable exception is the MCQA task where PCA performs very poorly,

which we suspect is due to the low-data regime of ≈ 100 examples harming the quality of the principal components we compute. This is in line with the results from the steering benchmark AxBench (Wu et al., 2025), in which SAEs struggle against simple steering baselines.

There is room for future submissions to establish state-of-the-art results on the benchmark. There are several clear opportunities for future submissions to the causal variable track to improve upon our baselines. First, our baselines provide weak evidence at best for the carry-the-one variable of the two digit addition task and we have hope that more complex feature spaces might be more successful. Second, while we only run baselines for IOI on the GPT-2 attention heads identified by (Wang et al., 2023), we encourage future submissions to locate and featurize attention heads in the other three models. Third, we did not conduct exhaustive hyperparameter searches for any of the baseline methods, so all tasks likely have room for improving the best alignment and average alignment across layers.

5. Related Work

Circuit discovery evaluation. While there do not exist benchmarks for circuit discovery methods in general, there do exist targeted tests of whether the *concept of* circuits is sensible (Shi et al., 2024). There are also benchmarks of circuit discovery methods in the domain where we have access to ground-truth circuits (Gupta et al., 2024), though the tasks in this benchmark are relatively simple, as the circuits are hand-crafted. Methods and metrics papers tend to focus on only one or two tasks and only one or two models (Miller et al., 2024; Ferrando & Voita, 2024; Conmy et al., 2023); this can function as a strong proof of concept, but limits our understanding of the generalizability and scalability of these methods and metrics.

Causal variable localization evaluations. The RAVEL (Huang et al., 2024a) and CausalGym benchmarks (Arora et al., 2024) both enable comparisons across featurization methods, though in more narrow domains. The SAEBench (Karvonen et al., 2025) is similar in concept, though much narrower in scope w.r.t. the kinds of methods that can be evaluated, i.e., only SAEs. Our benchmark compares across a range of tasks, models, and methods.

Other evaluations. There also exist benchmarks that do not fall cleanly within these two camps. An impactful application of MI methods is targeted model editing (Meng et al., 2022), for which there now exists multiple benchmarks (Cohen et al., 2024; Abraham et al., 2022; Zhong et al., 2023). An emerging paradigm is evaluating automated *interpretability agents*; for example, FIND (Schwettmann et al., 2023) evaluates the quality of interpretability agents in

correctly describing latent functions implemented by model components. Other benchmarks focusing on benchmarking explanations of LM behaviors (Mills et al., 2023; Atanasova et al., 2023).

6. Conclusions

We have proposed MIB, a Mechanistic Interpretability Benchmark, and demonstrated its value for directly comparing mechanistic interpretability methods. MIB corroborates recent findings, like the value of attribution methods, and challenges others, like the utility of SAEs as featurizers for known causal variables. MIB is not in its final form: as progress in MI is rapid, we intend this as a **living benchmark** that scales to incorporate new advances in the field. We therefore intend to support MIB such that our formulation can support future state-of-the-art methods.

Acknowledgements

We are grateful for feedback and ideas discussed with Neel Nanda, Sandro Pezzelle, Christopher Potts, and Tamar Rott Shaham in an early phase of this project.

This research was supported by a postdoctoral fellowship under the Zuckerman STEM Leadership Program (A.M.), grants from Open Philanthropy (A.G., Y.B., N.P., D.B.), grants from AI Safety Support Ltd (I.A., R.G.), the OpenAI Superalignment Fast Grant Program (M.H.), the Ariane de Rothschild Women Doctoral Program (D.A.), an armasuisse CYD Doctoral Fellowship (A.St.), an Azrieli International Postdoctoral Fellowship (A.R.), an Azrieli Foundation Early Career Faculty Fellowship (Y.B.), a Google academic gift (Y.B.), the European Union (ERC, Control-LM, 101165402; Y.B.), and the Apple AIML PhD fellowship (H.O.). The opinions expressed here are those of the authors, and not of any of the mentioned organizations.

Impact Statement

This paper presents work whose goal is to standardize the evaluation of mechanistic interpretability methods. Advancements in interpretability methods will advance current approaches to AI safety and robustness, many of which rely on localization as part of their pipelines. This could also potentially result in better countermeasures against safety methods that are meant to increase harm. We do not anticipate that MIB will directly contribute to such harms; it will primarily allow researchers to come to stronger conclusions regarding which localization methods tend to be best *in general*, regardless of downstream application.